Data and how we measure

Three AI-assisted synthetic datasets, 200 rows each
Two inputs. Target: $y = 3x_1 + 2x_2$, without noise

κ	Hessian eigenvalues	Selected GD step	GD updates predicted = measured
1	1 and 1	1.000000	1
10	0.1 and 1	1.818182	75
100	0.01 and 1	1.980198	746

Start: (0, 0). Stop: full L2 gradient norm < 0.000001. Full-batch GD.
Preliminary measured results, reproduced. Distinct gradient points: 2 / 76 / 747.

Evaluation metrics

Updates to tolerance, distinct gradient points, final loss,
L2 gradient norm and L2 parameter error

Dataset check: κ = 1 ÷ 0.01 = 100

Prior work: MNIST and CIFAR-10 in Kingma and Ba (2015), Sections 6.1–6.3